

Package ‘bamp’

June 21, 2026

Type Package

Title Bayesian Age-Period-Cohort Modeling and Prediction

Version 2.2.0

Date 2026-06-06

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Description Bayesian Age-Period-Cohort Modeling and Prediction using efficient Markov Chain Monte Carlo Methods. This is the R version of the previous BAMP software as described in Volker Schmid and Leonhard Held (2007) <[DOI:10.18637/jss.v021.i08](https://doi.org/10.18637/jss.v021.i08)> Bayesian Age-Period-Cohort Modeling and Prediction - BAMP, Journal of Statistical Software 21:8. This package includes checks of convergence using Gelman's R.

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Encoding UTF-8

Depends R (>= 3.5.0)

Imports coda, graphics, parallel, stats, abind

LazyData true

VignetteBuilder knitr, R.rsp

Suggests knitr, rmarkdown, R.rsp, testthat (>= 3.0.0)

URL <https://volkerschmid.github.io/bamp/>,
<https://github.com/volkerschmid/bamp>

BugReports <https://github.com/volkerschmid/bamp/issues>

Config/roxygen2/version 8.0.0

Config/testthat/edition 3

NeedsCompilation yes

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apc	<i>apc S3 class</i>
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Description

Class for (Bayesian) age-period-cohort objects

Usage

apc()

Details

`bamp` will return an object of class `apc`. Available functions are

- `plot.apc` plots main effects
- `print.apc` print summary of model and effects
- `effects.apc` extract effects (mean, median and quantiles)

Value

apc class

apc.data	<i>Example dataset for APC model</i>
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Description

A dataset containing case counts and population numbers in eight age groups for ten years. Each age group consists of five years.

Usage

data(apc)

Format

- population: matrix of population data
- cases: matrix of case counts
- cov_p: covariate for period
- cov_c: covariate for cohort

apcSimulate

*Simulate from age-period-cohort model***Description**

This functions simulates a data set of cases on the Lexis diagram from given age, period and cohort effects. Population numbers have to be given; can be one number for all age group/period combinations.

Usage

```
apcSimulate(intercept, age, period, cohort, periods_per_agegroup, population)
```

Arguments

intercept	Intercept
age	Vector of effect for age groups
period	Vector of effects for periods
cohort	Vector of effect for cohorts
periods_per_agegroup	Periods per age group
population	Population number. Either a matrix or a scalar.

Value

List with number of cases (matrix) and population numbers (matrix).

See Also

```
vignette("simulation", package = "bamp")
```

Examples

```
age=sqrt(seq(5,0,length=10)); age<-1-age-mean(age)
period=15:1; period[8:15]<-8:15; period<-period/6; period<-period-mean(period)
periods_per_agegroup=5; number_of_cohorts <- periods_per_agegroup*(10-1)+15
cohort<-rep(0,60); cohort[1:10]<-10:1; cohort[41:60]<- -(1:20)/2; cohort<-cohort/10;
cohort<-cohort-mean(cohort)
simdata<-apcSimulate(-5, age, period, cohort, periods_per_agegroup, 1e6)
par(mfrow=c(3,1))
plot(age, type="l")
plot(period, type="l")
plot(cohort, type="l")
## Not run:
simmod <- bamp(cases = simdata$cases, population = simdata$population, age = "rw1",
period = "rw1", cohort = "rw1", periods_per_agegroup =periods_per_agegroup)
plot(simmod)

## End(Not run)
```

Description

Bayesian Age-Period-Cohort Modeling for the analyze of incidence or mortality data on the Lexis diagram. For each pixel in the Lexis diagram (that is for a specific age group and specific period) data must be available on the number of persons under risk (population number) and the number of disease cases (typically cancer incidence or mortality). A hierarchical model is assumed with a binomial model in the first-stage. As smoothing priors for the age, period and cohort parameters random walks of first and second order (RW1 or RW2) available. Deviance information criterion and effective number of parameters is computed for model comparison. Note that there is a non-identifiability in the likelihood of the APC-model, see e.g. Clayton and Schifflers (1987, DOI:10.1002/sim.4780060406), which indices some problems in interpreting the latent effects. Only for RW1 model, the parameters are (weakly) identifiable. Period and age groups do not need to be on the same grid, for example periods can be in one year intervals and age groups in five year intervals.

Additionally to the model described in Knorr-Held and Rainer (2001, DOI:10.1093/biostatistics/2.1.109), bamp can handle

- AP and AC models,
- models with and without global heterogeneity parameter (overdispersion),
- models with additional age, period and/or cohort heterogeneity,
- additional covariates.

Usage

```
bamp(
  cases,
  population,
  age,
  period,
  cohort,
  overdisp = FALSE,
  period_covariate = NULL,
  cohort_covariate = NULL,
  periods_per_agegroup,
  mcmc.options = list(number_of_iterations = 1e+05, burn_in = 50000, step = 50, tuning =
    500),
  hyperpar = list(age = c(1, 0.5), period = c(1, 5e-04), cohort = c(1, 5e-04), overdisp =
    c(1, 0.05)),
  dic = TRUE,
  parallel = TRUE,
  verbose = FALSE
)
```

Arguments

cases	number of cases
population	population number

age	prior for age groups ("rw1", "rw2", "rw1+het", "rw2+het", " ")
period	prior for periods ("rw1", "rw2", "rw1+het", "rw2+het", " ")
cohort	prior for cohorts ("rw1", "rw2", "rw1+het", "rw2+het", " ")
overdisp	logical, add overdispersion to model
period_covariate	covariate for period
cohort_covariate	covariate for cohort
periods_per_agegroup	periods per age group
mcmc.options	list of options for MCMC. • number_of_iterations: number of iterations per chain. • burn_in: number of iterations used as burnin at the beginning of the algorithm, these iterations will be removed. • step: Step size, for example default is 50, so only every 50th iterations will be stored. • tuning: number of iterations for automatic tuning. Depending on the model, the MCMC algorithm will tune certain parameters for more efficient MCMC chains. After tuning, the algorithm is restarted.
hyperpar	list of hyper parameters. The hyper prior for the precision (inverse variance) in the random walk priors is a Gamma distribution with parameters a and b ; expected value is a/b , variance is a/b^2 . Weak hyper parameters are suggested, defaults are $a = 1, b = 0.5$ for age, $a = 1, b = 0.0005$ for period and cohort effects and $a = 1, b = 0.05$ for overdispersion (if added). It is recommended to choose the hyper priors depending on the model, in particular on the order of the random walk.
dic	logical. If true. DIC will be computed
parallel	logical, should computation be done in parallel. This uses the parallel package, which does not allow parallel computing under Windows.
verbose	verbose mode

Details

This functions returns an [apc](#) object. Only samples from the posterior are computed, point estimates and credible intervals will be computed in [effects.apc](#), [print.apc](#) and [plot.apc](#). [predict.apc](#) can be used for prediction of the future rates and number of cases and for a retrospective prediction for model checking.

See Also

```
vignette("modeling", package = "bamp")
```

Examples

```
## Not run:
data(apc)
model <- bamp(cases, population, age="rw1", period="rw1", cohort="rw1", periods_per_agegroup = 5)

## End(Not run)
```

checkConvergence	<i>Check apc object, whether MCMC has converged</i>
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Description

This functions uses Gelman and Rubin's R to check convergence for all main parameters. All parameters should have $R < 1.1$. `bamp` runs at least four MCMC chains by default (more if parallel is more than four).

Usage

```
checkConvergence(x, info = FALSE, level = 2, auto = FALSE)
```

Arguments

<code>x</code>	An apc object
<code>info</code>	logical; print more information
<code>level</code>	level of check; 1 uses point point estimation, 2 uses upper C.I.
<code>auto</code>	logical; should be TRUE if called automatically from <code>bamp</code> #'

Value

logical; TRUE if check is fine.

Examples

```
## Not run:
data(apc)
model <- bamp(cases, population, age="rw1", period="rw1", cohort="rw1", periods_per_agegroup = 5)
checkConvergence(model)

## End(Not run)
```

coh	<i>Compute cohort index from age and period index</i>
-----	---

Description

Compute cohort index from age and period index

Usage

```
coh(agegroup, period, noa, periods_per_agegroup)
```

Arguments

<code>agegroup</code>	age group index
<code>period</code>	period index
<code>noa</code>	number of age groups in total
<code>periods_per_agegroup</code>	periods per age group

Value

cohort index

Examples

```
# last agegroup in first period equals first cohort
coh(10, 1, 10, 5)

# first agegroup in last period equals last cohort
coh(1, 8, 10, 5)
```

effects.apc	<i>Effects from Fitted APC Model</i>
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Description

Effects from Fitted APC Model

Usage

```
## S3 method for class 'apc'
effects(object, mean = FALSE, quantiles = 0.5, update = FALSE, ...)
```

Arguments

object	an apc object
mean	logical. If TRUE, mean effects are computed
quantiles	Scalar or vector of quantiles to compute (only if mean=FALSE)
update	logical. If TRUE, the apc object including the effects is returned
...	Additional arguments will be ignored

Value

List of age, period, cohort effects or apc object including effects (if update=TRUE)

Examples

```
## Not run:
data(apc)
model <- bamp(cases, population, age="rw1", period="rw1", cohort="rw1", periods_per_agegroup = 5)
effects(model)

## End(Not run)
```

plot.apc	<i>Plot apc object</i>
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Description

Plot apc object

Usage

```
## S3 method for class 'apc'
plot(x, quantiles = c(0.05, 0.5, 0.95), ...)
```

Arguments

x	apc object
quantiles	quantiles to plot. Default: c(0.05, 0.5, 0.95) is median and 90% credible interval.
...	Additional arguments will be ignored

Details

Plot of age, period and cohort effects from apc objects. If covariates have been used for period/cohort, a second plot with covariate, absolute effect and relative effect is created. Absolute effect is relative effect times covariate.

Value

plot

Examples

```
## Not run:
data(apc)
model <- bamp(cases, population, age="rw1", period="rw1", cohort="rw1", periods_per_agegroup = 5)
plot(model)

## End(Not run)
```

predict_apc	<i>Prediction for age-period-cohort models</i>
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Description

Prediction of rates and, if possible, cases from the Bayesian age-period-cohort model using the prior assumptions (random walks) of the model and the estimated variance of the random walk. For example, random walk of first order (rw1) for period effect predicts constant effects for future periods plus noise.

Usage

```
predict_apc(
  object,
  periods = 0,
  population = NULL,
  quantiles = c(0.05, 0.5, 0.95),
  update = FALSE
)
```

Arguments

object	apc object
periods	number of periods to predict
population	matrix of (predicted) population, if NULL, population data from original bamp call will be used
quantiles	vector of quantiles to compute
update	boolean. If TRUE, object will be returned with results added to the object

Details

This function will return predicted rates for future periods. For this, future period and cohort effects will be predicted. Further age group effects will not be predicted. The rates are random samples from the predictive distribution; number of samples is equal to number of MCMC iterations. Quantiles will be provided for convenience, but all samples are available. If population numbers are given, number of cases will also be predicted. Number of cases will not only be predicted for future periods, but also for the time periods where data are available; this can be used for model assessment.

Value

list with quantiles of predicted probabilities (pr), predicted cases (cases) and predicted cases per period (cases_period) and a list samples with MCMC samples of pr, cases and cases_period. If update=TRUE, the apc object will be returned with this list (predicted) added.

See Also

```
vignette("prediction", package = "bamp")
```

Examples

```
## Not run:
data(apc)
model <- bamp(cases, population, age="rw1", period="rw1", cohort="rw1", periods_per_agegroup = 5)
pred <- predict_apc(model, periods=1)
plot(pred$pr[2,11,], main="Predicted rate per agegroup", ylab="p")

## End(Not run)
```

print.apc	<i>Print apc objects</i>
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Description

Print apc objects

Usage

```
## S3 method for class 'apc'  
print(x, ...)
```

Arguments

x	apc object
...	additional arguments will be ignored

Value

print

Examples

```
## Not run:  
data(apc)  
model <- bamp(cases, population, age="rw1", period="rw1", cohort="rw1", periods_per_agegroup = 5)  
print(model)  
  
## End(Not run)
```

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